# SAINT MICHAEL COLLEGE OF CARAGA

**THE EXTENT OF UTILIZATION OF MOBILE BANKING AMONG SENIOR HIGH STUDENTS**

A Thesis Proposal Presented To
The faculty of
College of Business Management and Accountancy

In Partial Requirements for the Degree
**Bachelor of Science in Business Administration**

By

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**SAINT MICHAEL COLLEGE OF CARAGA**
College Of Business Management And Accountancy

**CERTIFICATE OF RESEARCH APPROVAL**

This thesis entitled **"The Extent of Utilization of Mobile Banking Among Senior High Students"** presented and submitted by Daniella Maureen R. Caratao, Bhebiejoy L. Arquisola, Leriegine S. Felicilda, Carl Joven B. Olana, Jhaser Timothy S. Orihuela, and Vhea Mae R. Perez in partial fulfillment of the requirements for the degree of Bachelor of Science in Business Administration Major in Financial Management is hereby accepted and recommended for Oral Examination.

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**ABSTRACT**

| | |
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| **TITLE:** | THE EXTENT OF UTILIZATION OF MOBILE BANKING AMONG SENIOR HIGH STUDENTS |
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## ***RESEARCH DESIGN***

### I. OBJECTIVES

This study aims to assess the extent of mobile banking utilization among senior high students in Buenavista, Nasipit, and Carmen, Agusan del Norte. Specifically, determining which mobile banking features are most used, what benefits students perceive, and what barriers hinder their usage. This study also aims to identify strategies to enhance the adoption and effective use of mobile banking applications such as GCash and Maya among students. Finally, the study's findings will serve as the basis for developing a mobile banking literacy program for senior high school students.

### II. METHODOLOGY

This study will use a quantitative descriptive-correlational research design to assess the extent of mobile banking utilization among senior high school students in

Buenavista, Nasipit, and Carmen, Agusan del Norte. This method allows the researchers to describe the current level of usage of mobile banking applications such as GCash and Maya and to examine the relationship between demographic variables and the extent of utilization. The results of the study will serve as the basis for developing a literacy program to improve students' knowledge and responsible use of mobile banking services.

**III. FINDINGS**

**IV. RECOMMENDATIONS**

**KEYWORDS:** Mobile Banking, Mobile Payments, Senior High Students, GCash, Maya, and Financial Technology

## DEDICATION

## ACKNOWLEDGEMENTS

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# CHAPTER 1

# THE PROBLEM AND ITS SCOPE

## 1.1 Background of the study

Mobile banking has become a more common financial tool among Filipinos, particularly GCash, Maya, and other digital wallets, which enable users to manage their money easily on mobile devices. Mobile banking apps like GCash and Maya have become necessities for senior high school students, especially for allowances, online shopping, school transactions, and personal savings (Panganiban et al., 2022). However, among the senior high school students, many still do not fully utilize the mobile banking features. Other students use simple services like cash-in and payments. Still, some are afraid to do so due to a lack of financial literacy, fear of scams, poor internet access, or age constraints (Apau & Lallie, 2020). Despite these challenges, mobile banking continues to develop and is increasingly applied in schools and in daily student transactions. Therefore, the extent of mobile banking utilization is thus necessary in determining which mobile banking features are most used, what benefits students perceive, what barriers hinder their usage, and what strategies may help increase usage, and to use the findings as a basis for developing a literacy program.

The growing use of digital platforms has seen the emergence of mobile banking, where more transactions on goods and services are being made on the internet via mobile payments, e-wallets, and electronic payment systems (Agur et al., 2020; Alkhowaiter, 2020). Mobile banking applications often offer tools such as budgeting and expense

tracking, which help students manage their finances and improve their financial practices (Linawati & Wijaya, 2022). Thirunavukkarasu and Krishnamoorthy (2024) emphasize that understanding the challenges and facilitators influencing mobile banking adoption among students is crucial for improving usability, expanding the merchant network, and boosting e-wallet literacy. However, poor internet connection, lack of digital and financial literacy, security concerns, problems with technology, parental restrictions, and a preference for cash have caused students to limit the use of mobile (Apau & Lallie, 2020; Revana & Suhartini, 2024; Zeya, 2022; AlAli & AlAli, 2020; Zainol, 2021). Similarly, a study by Hussein et al. (2020) researched investment adoption in Pakistan, the results indicate that, despite the potential of mobile banking to enhance financial inclusion, low levels of digital literacy, mistrust of mobile platforms, and a poor understanding of mobile financial services are among the barriers to its adoption. This study suggests establishing effective training programs to enhance digital literacy and adoption.

The Philippines was not exempt from mobile banking adoption, as it is considered one of Southeast Asia's highest digital payment transaction countries in 2022 (Statista Research Department, 2022). Mobile wallets come in the form of Shopee Pay, GCash, Maya, and Grab Pay, among others. The widespread use of these mobile wallets prompted the Bangko Sentral ng Pilipinas (BSP) to respond by issuing Circular 702, which provides consumer protection, transparency, and information disclosure guidelines for banking services, including electronic and mobile banking. The top factors affecting students' utilization of electronic wallets were speed and convenience because regular monetary support/allowance from their parents and guardians is their main reason for

using them (Panganiban et al., 2022). Moreover, students' simultaneous use of electronic wallets is because of the following advantages: time efficiency, security, benefits, and promotions (Khan et al., 2020). For students, time efficiency in managing financial transactions allows more focus on academic pursuits, while vigorous security features safeguard their limited funds. Additionally, benefits such as promotions provide opportunities to save money or earn rewards, all contributing to a more financially stable student experience. In addition, the convenience and accessibility of digital payment platforms have also penetrated the education sector. As universities adopted alternative learning methods, electronic payments became essential for enrollment fees and related academic expenses (Espeleta, 2022). Philippine studies have examined the age, income level, education, and occupation of people, and how these factors affect the younger generation's use of mobile banking. However, the findings of the researchers were not the same (Belmonte, 2024; Panganiban et al., 2022; Alcain et al., 2024). Lastly, Panganiban et al. (2022) state that students' utilization of mobile banking applications is strongly influenced by convenience, transaction speed, and internet connectivity, and they point out the importance of efficient and reliable digital payment systems for young users.

Despite significant research on mobile banking in other developing countries, there is limited literature in the Philippine context that focuses specifically on senior high school students as new digital financial users. This gap is particularly evident in the municipalities of Buenavista, Nasipit, and Carmen in Agusan del Norte, where, to the best of the researcher's knowledge, no documented studies have specifically examined the extent of mobile banking utilization among senior high school students. This study

addresses these gaps by assessing the actual extent of mobile banking usage in relation to demographic variables, identifying what features are used, related barriers and perceived benefits, and proposing evidence-based strategies to enhance mobile banking utilization among senior high school students and to design a literacy program that improves students' digital financial literacy and encourages a healthy, secure, and inclusive use of mobile banking.

As a result, this study bridges a gap in local knowledge about how senior high school students utilize mobile banking and what factors influence their usage. The results can help various stakeholders, such as students, enhance their financial literacy, conduct safer online transactions, and increase awareness of mobile banking features. Department and school administrators will serve as a basis for creating an intervention program that effectively improves students' digital and financial awareness and reinforces responsible mobile banking use at schools. Mobile banking providers may also benefit by identifying users' needs and challenges to inform the development of youth-friendly services. Lastly, policymakers and community stakeholders, the results can inform programs promoting digital financial literacy among the youth.

**1.2 Review of Related Literature**

The adoption of mobile banking is primarily influenced by the positive user experience they provide. According to Handarkho & Harjoseputro (2019), the pleasure and enjoyment that new technology offers can significantly drive consumer adoption. Khan et al. (2020) found that younger generations, especially students, favor e-wallets

because they are convenient. Digital payment encompasses transactions conducted via electronic devices and platforms (Dimitrova & Yazdanfar, 2021). The targeted services of various studies are these terminologies; while they differ slightly in features (e.g., money transaction), they combine to form the family of m-banking services. One of the most popular m-banking services is mobile payment.

Furthermore, multiple factors influence the adoption and extent of utilization of these platforms (Gcash, Maya, Others). The two factors that the users most frequently identify are perceived usefulness and ease of use, whereas digital literacy, service awareness, and socio-demographic characteristics such as age, income and level of education are also significant contributors to the same effect (Diaz, 2023; Velez, 2025). Similarly, with the rise of digitalization and cashless transactions, a study in Indonesia found that infrastructure, income level, and education significantly influence individuals' perceived readiness to adopt a cashless lifestyle (Balakrishnan et al., 2021). Lastly, a study by Ervas (2025) analyzed four variables, such as features, benefits, barriers, and strategies, to effectively evaluate the level of mobile banking utilization in Gubat, Sorsogon, and to come up with a recommendation on how financial inclusion can be enhanced. It pinpoints the features those users actually use, the benefits they value, and the barriers they encounter—information that previous studies did not clearly detail.

**Mobile Banking Features**

Mobile banking features vary in nature, which directly affects user adoption and the intensity of use. Features of mobile banking are growing, i.e., sending and receiving money, paying bills, contactless shopping, and so on (Zeya, 2022). The various mobile

banking services/features also include payment, shopping, and fund transfer. The development of this technology has significantly increased students' satisfaction by enhancing convenience, expanding the variety of mobile application features, simplifying usability, and offering greater flexibility—critical attributes in financial services that drive higher customer satisfaction, as highlighted by Zhang and Kim (2020).

Furthermore, the high percentage is explained by the increasing trend towards the use of electronic channels when it involves fulfilling periodical financial obligations like an electricity payment, a mobile payment, and other routine financial commitments (Goveas et al., 2025). In the Philippines, Ervas (2025) found that fund transfers, ranked first, are the most regularly used mobile banking feature. Darwin proposes that users utilize mobile banking more frequently for traditional financial operations, such as transferring money between accounts and monitoring balances, instead of using features like investments that may be challenging for less financially literate or trustworthy individuals.

**Perceived Benefits**

The previous studies on mobile banking and digital payments observed that the perceived benefits/relative advantage have a significant positive influence on mobile banking adoption. Perceived benefits refer to the users' evaluation of the advantages of using mobile banking services. Davis (1989), in the Technology Acceptance Model (TAM), explained that perceived usefulness is a primary factor influencing technology adoption. Perceived benefits are defined as the more frequent use of an innovation as compared to its existing alternatives (Jung et al., 2020). Mobile banking provides relative benefits compared to debit cards, ATMs, and internet banking. Mobile banking operations can be

performed round the clock and round the globe and do not have any spatial or temporal limits, like internet banking. Initiating and conducting financial transactions through mobile banking is more convenient, efficient, and inexpensive. Furthermore, Mombeuil (2020) found that perceived benefit has a significant positive influence on mobile wallet adoption.

Several studies, including those by Booranapim et al. (2020), have shown that perceived benefits significantly influence the intention to use mobile banking. The organization could offer users benefits from using the application and provide additional incentives to attract more users. Lastly, Ervas (2025) reports that mobile banking benefits provide more convenient, affordable, and quick financial services. It allows customers to conduct transactions 24/7.

**Barriers**

Previous research by Khan et al. (2020) and Nasir et al. (2021) identified significant barriers to mobile banking. The main factors behind these challenges are a lack of literacy and trust in mobile banking practices. The elderly group is especially affected by this drawback, as they have, by nature, lower literacy and may not see the worth that mobile banking services offer. These findings support those of Błach & Klimontowicz (2021), Aziz & Naima (2021), and Behera et al. (2023), who showed the impact of network problems and poor connectivity on the uptake of digital payments.

Students experience various barriers when utilizing mobile banking services. Among them, poor internet connectivity is a major factor in use (Panganiban et al., 2022), and low digital or financial literacy, which only provides students with basic features (Silva

et al., 2024). Security concerns, such as fraud and phishing, discourage adoption (Apau & Lallie, 2020). Technical problems and errors in the app are additional barriers (AlAli & Alali, 2020), as are parental restrictions (Panganiban et al., 2022).

According to Goveas et al. (2025), the adoption of mobile banking apps and their subsequent usage has been growing tremendously in the recent past due to technological progress and shifting consumer preferences. Despite this progress, several barriers prevent customers from using these apps more frequently, including a lack of trust in digital transactions and mobile banking. The top-ranking barriers include a lack of awareness and a lack of internet connection, both of which are tremendous deterrents to mobile banking use. Several studies, including those by Ervas (2025), have identified several barriers that impede the use of mobile banking. Low levels of digital literacy make a significant number of people hesitant to use mobile applications, and there is a low internet connection, which influences the availability of services and their quality.

**Proposed Strategies**

Panganiban et al. (2022) examined the utilization of mobile money applications among business students and proposed several strategies to improve the use of digital financial platforms. They recommend conducting awareness and education campaigns on the convenience and speed of mobile money, additional security education to address user concerns, and partnerships with mobile money providers to develop and implement features and incentives that appeal to students. They further emphasized the need to strengthen internet connectivity and to put in place a continuous monitoring and feedback system to improve mobile money services gradually. Similarly, Aboagye and

Boateng (2022) found that education workshops strengthen the ownership of mobile money accounts, increase the frequency of mobile money transactions, and raise the proportion of financial transactions conducted using mobile money.

Additionally, Hussein et al. (2020) conducted a study that investigated investment adoption in Pakistan. The findings also show that while mobile banking has the potential to increase financial inclusion, several barriers hinder its adoption. Some of the factors include low levels of digital literacy, mistrust in mobile platforms, and a poor understanding of mobile financial services. The study recommends developing efficient training programs to improve digital literacy and adoption. Lastly, according to Ervas' (2025) findings, several solutions to these problems are offered, including community-based digital literacy programs to inform users, easier internet access by collaborating with telecom companies, and simpler mobile banking interfaces. A combination of these measures is used to increase mobile banking use and advance financial inclusion for communities.

**Senior High School Students**

Senior high school students are a distinct population group in research due to their transitional developmental stage and high engagement with digital technologies. Various studies indicate that most senior high school students are highly exposed to mobile devices and the internet for educational and personal purposes (Wan Jin & Junio-Sabio, 2025). Research also highlights varying levels of digital literacy among senior high school students, which can influence their interactions with digital tools and services and, in turn, shape these interactions (Barri et al., 2025). Additionally, the research on financial literacy

in the group reveals that knowledge of financial behavior is essential to the program to enhance financial decision-making (Nicole et al., 2024). These findings align with broader evidence that youth in this developmental stage are already active participants in digital financial ecosystems but still need tailored education to take full benefit of the technologies (OECD, 2020).

Furthermore, the top factors affecting senior high school students' utilization of mobile wallets were speed and convenience, with regular parental or guardian financial support or allowance as their main reason for using them (Panganiban et al., 2022). Lastly, earlier research has indicated that a majority of senior high school students use digital financial services for their daily allowances, online shopping, and school-related transactions. Hence, this demographic is appropriate for studying the use of mobile banking and the factors influencing its utilization (UNICEF, 2021; Panganiban et al., 2022).

**Age**

Regarding age, older people are more likely to adopt mobile banking, whereas Mori & Mlambiti (2020) found that age was not a significant predictor of adoption. The author found a positive relationship between adopting digital banking because of the accessibility of Internet services and improved infrastructure. The older they become, the more likely they are to embrace digital banking. However, age is negatively associated with the probability of online banking adoption. Consumers are less likely to adopt electronic banking, according to Yates (2020). The rate of older consumers remains lower than that of younger consumers, despite the increasing adoption rates for digital banks over time.

Moreover, age is negatively associated with the probability of online banking adoption. Consumers are less likely to adopt electronic banking. The author also noted that most systems have failed because, despite the rising adoption of digital banking, customers are not using the technology sufficiently to justify the costs of development and maintenance, particularly among those unfamiliar with technological advancements. According to Lachhwani & Kanwar (2020), older people still do not prefer to use digital banking, as they think traditional banking is more convenient. This study also implies and discusses that older people believe banks must follow some strategic approaches to satisfy customers' changing demands when adopting digital banking.

However, some studies identified no statistically significant age-moderating effect on the relationship between financial literacy and mobile banking usage. In other words, while age might vary between users, it did not significantly affect the extent of mobile banking app usage (Sunday Adewale Olaleye et al., 2022). Lastly, in terms of age, research has shown significant variations in the adoption of mobile banking services.

**Sex**

According to Vinitha. k (2021), sex is a vital predictor of behavioral intention in mobile banking. Still, sex does not have any influence on attitudes toward the use of digital banking. This study highlights that sex is very crucial when it comes to the use of mobile banking, specifically in behavioral intention. However, Sandrasegaran and Rambeli (2023) also found that female students save more frequently and that parental income significantly shapes student spending habits.

Moreover, empirical findings and an improved understanding of the factors influencing consumer acceptance and actual use, as well as how they differ between developing and established nations, can be obtained by examining sex as a moderator of mobile banking adoption (Merhi, 2021). The researcher noted that sex can influence the adoption of digital banking in different ways, depending on a country's level of development. By adding sex as, the moderator, the study will provide an exact understanding of consumer adoption and its actual use (Christian et al., 2021; Sharma et al., 2022; Vinitha. k, 2021).

On the other hand, sex was found to have no apparent effect on the adoption of digital banking when demographic characteristics were controlled for (Alnemer, 2022). The research results indicate that both men and women are more likely to use digital banking, regardless of whether it is hindered. This study also shows that adopting digital banking is not hindered by sex. However, they reveal an intriguing pattern, suggesting that sex may not be essential for assessing the adoption of mobile banking due to its minimal impact (Mori & Mlambiti, 2019). This study implies that while sex was unimportant in adopting, it still presents an exciting trend. Furthermore, this study indicated that sex may not have an overall impact, but it still influences the adoption patterns in different contexts.

**Income (Monthly Allowance)**

Income significantly influences mobile banking utilization among senior high school students by affecting access to smartphones, data acquisition for mobile use, and the availability of sufficient funds for e-transactions. Many studies have shown that

individuals with higher incomes have a higher tendency to adopt mobile banking services. This can be evidenced by the study done by Abayomi et al. (2019) in Nigeria, where income positively influenced mobile banking use.

Revana and Suhartini (2024) discovered that students with better income (allowance) are using mobile banking to a greater extent, and Zeya (2022) found that low-income young people use only the basic features and do not refer to more complicated digital financial services. It was also noted by Panganiban et al. (2022) that mobile money application usage depends on the family income of students. Overall, income is a motivational determinant and the capacity to use mobile banking, and that is why it is a vital demographic factor of student adoption (Revana & Suhartini, 2024; Zeya, 2022; Panganiban et al., 2022). However, Zeya (2022) found that income does not significantly influence mobile banking adoption during the COVID-19 pandemic.

**GCash**

According Rakuten Insight (2022), 98 percent of respondents in the Philippines reported using GCash for their e-payment transactions, making it the country's largest e-payment service or e-wallet. Based on prior research, customers prefer cashless payment methods over traditional payment methods because they find them more comfortable and easier to use (Abdullah A. et al., 2021). GCash and Maya provide financial services to every Filipino, even those without access to a physical bank. Besides, they have enabled users to make purchases in-store, thereby greatly increasing their use and acceptance among Filipinos and guaranteeing their dominance in the financial services sector (Abdullah et al., 2021).

The Philippine Economic Society (2020) found that GCash has enabled more people to participate in the digital economy. It has also helped to reduce the cost of financial transactions and increase access to financial services for the unbanked population. GCash and Maya are options for Filipinos to do digital transactions without physical cash and credit cards (Zoleta, 2021). GCash remains the most popular app on the Apple App Store for its unique features and convenience (Zoleta, 2021). Espeleta (2022) found that most students use GCash more than any other mobile payment app. Students pick GCash because it's easy to access, they already know how it works, and many schools and businesses accept it. On the other hand, PayMaya doesn't catch on as much—mainly because fewer people use it, and there's less trust in the platform.

**Maya**

Maya, formerly known as PayMaya, is one of the leading mobile banking and e-wallet platforms in the Philippines that provides mobile services for fund transfers, QR payments, bill payments, online purchases, credits, and savings. PayMaya is appealing for its low ATM fees as the market leaders in the Philippines' digital payment space, GCash and Maya offer a wide range of services designed to meet customer preferences (Kshirsagar & Rai, 2022).

GCash and Maya are options for Filipinos to do digital transactions without physical cash and credit cards (Zoleta, 2021). Furthermore, GCash and Maya provide financial services to every Filipino, even those without access to a physical bank. Besides, they have enabled users to purchase in-store, thereby greatly increasing their use and acceptance among Filipinos, thereby guaranteeing their dominance in the financial

services sector (Abdullah et al., 2021). Cabrera et al. (2023) found that among accounting students, only a small proportion use PayMaya, which GCash and ShopeePay overtake in terms of overall usage. It is most widely used as a backup or substitute e-wallet, with a limited number of students making use of it. While PayMaya still works for some transactions, its limited usage correlates with lower merchant adoption and a less pronounced preference compared to other digital wallets.

**Literacy Program**

Literacy programs, particularly financial and digital financial literacy, are critical to empower people to cope with the modern economy. Literacy education is essential for the knowledge and skills students need to make informed financial decisions, such as budgeting, saving, managing credit, and investing. Research also indicates that financial literacy is positively correlated with good financial behavior and financial inclusion, especially in developing countries, where the challenges facing financial services are even more apparent (Islam & Khan, 2024).

Furthermore, the literature defines a Financial Literacy program as the knowledge and skills required to access and utilize mobile banking for financial services or to purchase goods and services (Mmari et al., 2024). Digital financial literacy has become an essential skill for managing personal finances, especially as students increasingly use digital financial tools (Silva et al., 2024). A well-rounded education in digital financial literacy is essential for minimizing risk while empowering students to make financial decisions on their own (Dheepiga, S., 2024).

According to the literature, there is a combination of technological, personal, and contextual factors that determine the adoption of mobile banking. Platforms like GCash and Maya are highly favored for their convenience, ease of access, and their feature sets, which include fund transfers and bill payments. These definitely cater to students and younger users who are heavily into digital apps.

However, despite the increasing adoption of mobile banking, some barriers limit its effectiveness. There are multiple problems, such as a lack of digital and financial literacy, security and privacy, unstable access to the internet, and a lack of awareness of its advanced features. Senior high school students face these problems very intensely because, along with parental restrictions, they have to manage with inconsistent allowances and limited access to digital resources. Besides that, demographic variables such as age, sex, and income (monthly allowance) further influence the extent to which students utilize mobile banking services. However, findings across studies indicate that their influence may vary depending on context.

The reviewed studies highlight the importance of financial and digital literacy programs in overcoming challenges related to mobile banking. Research indicates that these programs improve users' knowledge, confidence, and responsible use of mobile banking platforms, thereby promoting financial inclusion.

**Theoretical Framework**

The Unified Theory of Acceptance and Use of Technology (UTAUT) by Venkatesh et al. (2003) and the Diffusion of Innovation (DOI) Theory by Everett Rogers (2003) are the two theories underpinning this study, as they are complementary to each other. Therefore, these theories collectively explain both the behavioral and social processes that influence the extent of mobile banking utilization among senior high students and guide the development of intervention program that enhance financial inclusion in Buenavista, Nasipit, and Carmen, Agusan del Norte.

The quantitative side of the study is based on the UTAUT model, which can be used to explain individuals' acceptance and usage of mobile banking technology. It assumes that there are four significant factors of technology use, namely Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI), and Facilitating Conditions (FC). These constructs correspond directly to the study's dependent variable—the Extent of Utilization of Mobile Banking. Particularly, performance expectancy is associated with the perceived benefits of mobile banking, like convenience, speed, and efficiency in carrying out financial operations. Effort expectancy is linked to perceived ease of use of mobile banking applications such as GCash and Maya, which means how easy and convenient these systems are to use among rural consumers. Social influence refers to the support or validation from colleagues, relatives, or retailers that affects the likelihood of using mobile banking. Lastly, facilitating conditions refer to the external resources, including a constant internet connection, mobile access, and user support, without which it is impossible to adopt and further use the technology.

In the meantime, the demographic profile of the respondents, specifically age, gender, income, and the mobile banking platforms that they utilize, may be viewed as a moderating factor in the UTAUT model. These factors influence how individuals perceive mobile banking and their reactions to its features, benefits, and barriers. Younger, more educated users tend to exert greater effort and set higher performance expectations. Similarly, the preferred mobile banking platforms, each offering varying levels of accessibility, ease of use, and security, can influence the user experience. These personal differences contribute to variations in the extent of mobile banking utilization among senior high school students.

In addition to UTAUT, the Diffusion of Innovation (DOI) Theory by Rogers supports the quantitative aspect of the study since it describes how mobile banking spreads and is adopted among senior high school students. According to the DOI theory, the adoption of innovations like mobile banking occurs in five phases: knowledge, persuasion, decision, implementation, and confirmation. In the context of this study, DOI helps interpret the extent of utilization by demonstrating how age, gender, income, and preferred mobile banking platforms influence students' movement through these stages.

By integrating UTAUT and DOI, the study captures both the extent (how often and how much mobile banking is used) and the process (how and why it is adopted) of mobile banking use. UTAUT helps create attribute measurement variables, such as the relationship between demographic profiles and utilization levels. Meanwhile, DOI complements this by explaining how and why students' progress through the stages of adopting mobile banking—from initial awareness to regular, consistent use. Combining

these theories explains the behavioral, technological, and social forces that influence the extent of mobile banking utilization and provides a strong theoretical foundation for interpreting the study's quantitative findings.

**Schematic Diagram**

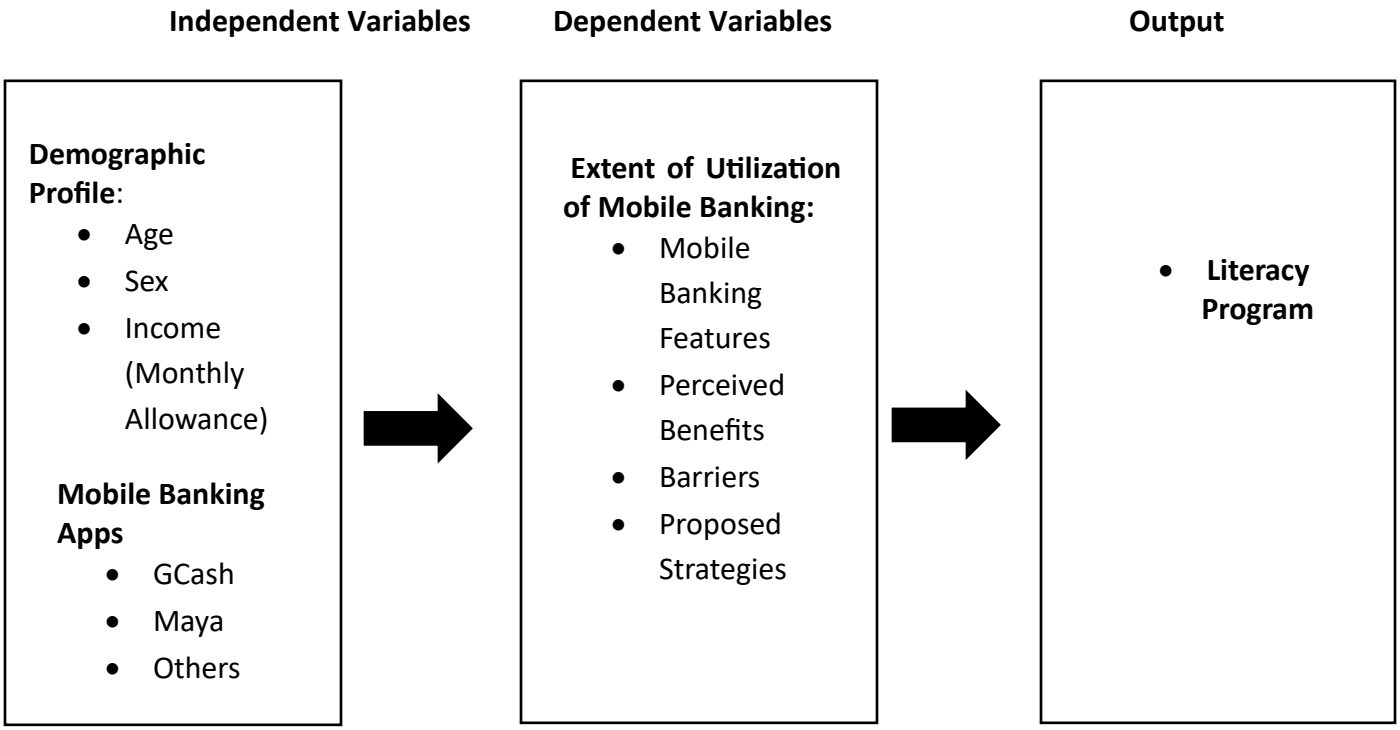



**Fig. 1** The schematic diagram shows the interplay between the independent variables (demographic profile), the dependent variable (extent of utilization of mobile banking), and the output (literacy program) that reflect the overall framework of the study.

**Statement of the Problem/Objectives of the Study**

The primary purpose of this study is to assess the extent of mobile banking utilization among senior high school students in Buenavista, Nasipit, and Carmen, Agusan del Norte, and to use the findings as a basis for developing a literacy program that may enhance students' understanding, awareness, and safe usage of mobile banking

Specifically, it aims to answer the following questions:

1. What is the demographic profile of the Respondents in terms of:

    1.1. Age;

    1.2. Sex;

    1.3 Income (monthly allowance); and

    1.4 Mobile Banking Platform used?

    1.4.1 Gcash;

    1.4.2 Maya; and

    1.4.3 Others?

2. What is the extent of utilization of mobile banking among the senior high students in terms of:

    2.1 Mobile Banking Features;

    2.2 Perceived Benefits;

    2.3 Barriers; and

    2.4 Proposed Strategies?

3. Is there a significant relationship between the respondent’s demographic profile and their extent of mobile banking utilization?

4. Based on the findings, what mobile banking literacy program may be proposed to enhance mobile banking utilization among senior high school students?

**Hypothesis**

**Null Hypothesis ($H_o$)**

There is no significant relationship between the respondents' demographic profile (age, sex, income (monthly allowance), and mobile banking platforms used) and the extent of mobile banking utilization among senior high school students.

**Scope and Delimitations**

This study focuses on assessing the extent of mobile banking utilization among senior high school students in the municipalities of Buenavista, Nasipit, and Carmen, Agusan del Norte. The selected locations represent urban, semi-urban, and developing communities, where student access to mobile banking platforms such as GCash and Maya is increasingly prevalent. It specifically examines how students use mobile banking applications such as GCash, Maya, and other digital banking platforms and evaluates their utilization in terms of their features, perceived benefits, barriers, and proposed strategies. The study also examines how demographic variables—such as age, sex, income (monthly allowance), and the mobile banking platforms used—affect the extent of utilization. The findings of this study will serve as a basis for developing a literacy program that may enhance the mobile banking awareness, safety, and usage among senior high students.

The study utilizes a quantitative descriptive-correlational design, as the primary data-gathering instrument. The study established these delimitations to ensure the research remained focused and manageable while achieving its objectives. Therefore, the

results of this study cannot be generalized to students at other levels of education or to users of mobile banking platforms beyond this study.

**Limitations of the Study**

Despite careful planning, the research has a few limitations. To begin with, the information relies on self-reports, which are likely to be influenced by response bias or, conversely, inaccurate. Second, respondents' availability during school hours may limit data collection. Third, time and resource constraints narrowed the research to specific schools, preventing a more comprehensive or longitudinal analysis. Lastly, disparities in internet access and digital familiarity among respondents may have influenced their experiences and responses to mobile banking.

**Definition of Terms**

The following terms were operationally defined as:

**Age.** It pertains to the respondents' current year of existence.

**Barriers**. It refers to the identifiable factors that hinder or limit the adoption, access, and regular use of mobile banking services.

**Income (Monthly Allowance).** It refers to the estimated monthly allowance or personal income the respondents received from parents or guardians.

**Literacy Program.** It refers to a structured educational plan that will be developed based on the study's results to enhance senior high school students’ knowledge, awareness, and safe use of mobile banking applications.

**Mobile Banking.** It refers to the delivery of financial products and transactions via mobile devices (such as smartphones or feature phones) on digital platforms like GCash and Maya, allowing users to transact without visiting physical bank branches.

**Mobile Banking Features.** It refers to the specific, interactive functionalities of mobile applications that enable users to conduct financial transactions and access services.

**Perceived Benefits.** It refers to the users' personal assessment of the advantages and positive outcomes they gain from using mobile banking platforms such as GCash and PayMaya in their daily financial transactions.

**Proposed Strategies.** It refers to the practical, actionable suggestions that have been identified in the study, which are meant to enhance the banking awareness, safety, and utilization of mobile banking among senior high school students.

**Sex.** It pertains to the respondents' identity.

**Utilization.** It refers to the extent to which senior high school students actually make use of mobile banking applications for their financial transactions, including frequency of use and the different features they engage with.

# CHAPTER 2

# METHODOLOGY

This chapter discussed and dealt with what researchers adhered to answer the specific problems posed by systematic investigation. Specifically, the research design, locale, research instrument, ethical standards, data gathering procedure, and statistical treatments will be used.

## 2.1 Research Design

This study will use a quantitative descriptive—correlational research design to assess the extent of mobile banking utilization among senior high school students in Buenavista, Nasipit, and Carmen, Agusan del Norte. This design is appropriate because it allows for the systematic description of the extent of mobile banking utilization among senior high school students and the examination of relationships between respondents' demographic profile and their level of utilization using statistical methods.

The descriptive component assesses respondents' demographic profiles, specifically their age, gender, income (monthly allowance), and the mobile banking platforms they use, while also measuring the extent of mobile banking utilization regarding features, perceived benefits, barriers, and proposed strategies. Meanwhile, the correlational component will be used to examine whether there is a significant relationship between the demographic profile of the respondents and their extent of mobile banking utilization. A structured survey questionnaire will be used to gather data, appropriate for collecting numerical data that could be analyzed statistically. The results

of the quantitative findings will serve as the basis for the development of a literacy program, which is the final output of the study.

**2.2 Research Locale**

The study will be conducted in the municipalities of Buenavista, Nasipit, and Carmen in the province of Agusan del Norte, Philippines. These municipalities include many public and private schools that provide a senior high school program, making them a suitable context for assessing the extent of utilization of mobile banking among senior high school students.

Buenavista has a land area of 475.61 square kilometers and a total population of 68,892, according to the 2020 census, a first-class municipality with a growing population and educational facilities catering to the urban and semi-urban populations. Nasipit has 144.40 square kilometers and a total population of 44,822, as determined by the 2020 census, a third-class municipality and home of the province's main seaport, has a diverse student population, and has growing access to digital technologies. Lastly, the municipality of Carmen has a land area of 311.02 square kilometers and a total population of 23,172, as determined by the 2020 census, classified as a third-class municipality as well, is composed of various barangays, where senior high school serves as a pathway for students from different socioeconomic backgrounds.

These locations were selected because they collectively represent a range of community types, i.e., urban, semi-urban, and developing communities, where mobile banking services like GCash and Maya are becoming more popular among students for school-related and personal transactions. Conducting the study in these municipalities

allows for a broader understanding of how demographic factors influence the extent of mobile banking utilization among senior high school students in Agusan del Norte.

**2.3 Research Respondents**

***Table 1. Distribution Of Respondents***

| Area / School | Grade Level | Population (F) | Sample Size (f) | Percentage (%) |
|---|---|---|---|---|
| **Buenavista** | | | | |
| Buenavista Institute | Grade 11 | 74 | 9 | 2.7 |
| | Grade 12 | 99 | 12 | 3.6 |
| Saint James | Grade 11 | 421 | 51 | 15.1 |
| | Grade 12 | 357 | 43 | 12.7 |
| **Nasipit** | | | | |
| Saint Michael College of Caraga | Grade 11 | 278 | 34 | 10.1 |
| | Grade 12 | 264 | 32 | 9.5 |
| Nasipit National Vocational School | Grade 11 | 295 | 36 | 10.7 |
| | Grade 12 | 270 | 32 | 9.5 |
| Calixto D. Montinola National High School | Grade 11 | 72 | 9 | 2.7 |
| | Grade 12 | 80 | 10 | 3.0 |
| **Carmen** | | | | |
| Our Lady of Carmen Academy of Caraga, Inc. | Grade 11 | 62 | 8 | 2.4 |
| | Grade 12 | 70 | 9 | 2.7 |
| Senior High School in Carmen | Grade 11 | 213 | 25 | 7.4 |
| | Grade 12 | 227 | 28 | 8.3 |
| **Total** | | **2,782** | **338** | **100** |

Table 1 shows the distribution of the respondents in the municipalities of Buenavista, Nasipit, and Carmen, Agusan del Norte, based on their respective schools and grade levels. The data show that the respondents consist of Grade 11 and Grade 12 senior

high school students who enrolled in different public and private secondary schools in the three municipalities.

The total senior high school population across these schools is 2,782 students. Using the Cochran's Formula to determine sample size at a 95% confidence level, the calculated sample size is 338 respondents. The following criteria were used to include the respondents: (1) the student must currently be enrolled as a senior high school student in the selected schools located in Buenavista, Nasipit and Carmen, Agusan del Norte; (2) the student must have experience with or be familiar with mobile banking applications; (3) the student must be willing to participate in the study and provide informed consent, with parental assent required for those under 18 years of age; and (4) the student must be present during the data collection period. This sample, proportionally distributed across each school and grade level based on actual enrollment figures, ensures comprehensive representation of the entire senior high school population.

The participating schools include Buenavista Institute, Saint James, Saint Michael College of Caraga, Nasipit National Vocational School, Calixto D. Montinola National High School, Our Lady of Carmen Academy of Caraga, Inc., and the Senior High School in Carmen. Both Grade 11 and Grade 12 students were included, and the study will be able to capture the usage patterns of the various year levels.

A stratified random sampling method was applied, wherein each school and the grade levels were taken as individual strata. The number of respondents selected from each stratum was proportionate to its population size. This method provides equal representation in the schools and municipalities and increases the reliability of the

findings regarding the extent of mobile banking utilization among the senior high school students.

**2.4 Research Instruments**

The research will use a structured survey questionnaire as the primary instrument for data collection. The survey questionnaire will be modified and adapted from Darwin F. Ervas's (2025), "The Extent of Utilization of Mobile Banking in Rural Community: Basis for Adoption to Strengthen Financial Inclusion."

The instrument is designed to gather data directly related to the Statement of the Problem, and therefore consists of two main parts, as follows:

**PART I – Demographic Profile of the Respondents**

This section will gather background information, including age, gender, income (monthly allowance), and the mobile banking platforms the user uses. These variables will help determine whether demographic factors influence the level of mobile banking utilization among rural residents.

**PART II – Extent of Utilization of Mobile Banking**

In this section, the researcher will quantify the frequency and efficiency with which respondents use mobile banking services (e.g., GCash, Maya, and others) to carry out various financial activities.

The responses will be rated using a four-point Likert scale, with the following range:

4 – Strongly Agree 3 – Agree 2 – Disagree 1 – Strongly Disagree

Content validation will be conducted on the survey questionnaire to assess clarity, relevance, and alignment with the study objectives. The research adviser will review and

validate the instrument by checking the items for appropriateness, clarity, and consistency with the study's variables. Prior validation, pilot testing, and reliability assessment were conducted by the original authors in an earlier study. Before the final distribution of the questionnaire, all modifications and suggestions will be put into practice.

**2.5 Research Ethical Standards**

**A. Intellectual Property Rights**

The study will not create patentable technologies, processes, or products, but will include original survey materials relevant to the data-gathering procedure. These will be protected under Philippine copyright law. Proper credit will be given to the secondary data and third-party sources for the information used in the research. There will be no unique brand, logo, or symbol in the study; hence, trademarks will be unnecessary.

**B. Informed Consent**

The researchers will thoroughly clarify the general objective of this study to the selected senior high school respondents from Buenavista, Nasipit, and Carmen who utilize mobile banking applications. This study involves Grade 11 and Grade 12 senior high school students, some of whom are minors. Before the study is conducted, official permission will be obtained from the school principals and teachers-in-charge of the participating schools. Before data collection, an orientation will be conducted explaining the purpose of the study, procedures, potential risks, and participants' rights. Students under 18 will receive an assent form for their parents or legal guardians to sign, along with an informed consent form for the student respondents. For the student to participate in the study, they must sign and approve both forms. Additionally, for respondents aged 18 and above,

informed consent forms will be provided and signed by participants before their participation. Participation in the study will be voluntary, and respondents will be told that they may withdraw at any time without penalty. Lastly, this study will have no ethical issues, as it will not use animals or other living things.

**C. Data Privacy and Confidentiality**

Participants' information, including their demographic details and responses, will be handled confidentially. Only the researchers involved in this study will have access to the personal information provided. The completed survey questionnaires will be kept in strict confidence, and access to them will be provided only to the researchers and the research adviser to ensure that the research data is handled ethically. After the data has been collected, the physical survey form will be securely stored in a sealed box for five (5) years to ensure its safety and integrity. This retention time allows the researchers to verify the results, accommodate future academic auditing, and comply with standard research archiving requirements. After five years, the physical instruments will be carefully shredded to prevent any unauthorized access to personal information. Digital files will also be permanently deleted from storage devices once the study is completed and the results have been finalized. This process ensures responsible data management and protects the rights and privacy of all respondents.

**D. Voluntary Participation**

The respondents will be fully informed of the purpose of the research survey and any potential risks it may entail. They will also know that they have the right to withdraw from the study at any time without incurring costs. To fulfill the ethical standards, they

will be assured about their personal information and anonymity. They will not be forced or bribed to be part of it; this will be of their own choice. The respondents will also be advised that they may step out at any time without adverse consequences.

**E. Minimization of Harm**

Respondents will be informed of the research's purpose, the procedures involved, potential risks, and their rights. Participation will be voluntary, with no forms of coercion. The respondents will be assured that their information will remain confidential and that their data will be protected against unauthorized access or breaches, especially personal or financial information. No respondents will be exposed to psychological, social, or financial harm during or after the research. Compensation will be provided to respondents if their privacy is compromised during or after the study. The survey questions and interview protocols will be carefully designed to avoid leading or manipulating respondents toward any particular viewpoint. Researchers will maintain transparency regarding the intended use of the research findings and the duration of data retention.

**F. Beneficence**

The research will contribute to the community, academics, and financial service providers by providing data-driven insights into the utilization of mobile banking among senior high school students. The results may also serve as a basis for the development of financial literacy programs, improving of digital payment services, and support for initiatives that encourage responsible and informed use of mobile banking. Moreover, the research findings will be available to respondents and other stakeholders upon request.

**G. Justice and Fair Participant Selection**

The study respondents will be selected using fair and transparent inclusion criteria. There will be no discrimination in the selection process; all individuals from diverse backgrounds who utilize mobile banking will be fairly included in the study.

**H. Data Integrity and Accuracy**

The researchers will complete their tasks with the best possible regard for the respondents' expectations. The researcher's primary ethical concern was protecting the respondent's privacy throughout the survey process. Researchers will safeguard the well-being and honesty of those who actively participate in the studies. To ensure the confidentiality of the information under consideration, the researchers must use the questionnaire directly.

**I. Transparency in Reporting**

There will be no conflicting interest with money or self-interest in the study. The researchers will analyze the data without input from mobile banking providers or other organizations. A potential conflict may arise; therefore, the details will be shared to enhance the study's credibility and integrity. Furthermore, the research findings will be presented honestly and transparently. The researchers will reference and acknowledge copyrighted and patented content to prevent conflicts and to respect intellectual property rights. Additionally, the researchers will utilize AI to help advance the study.

**J. Use of Patented or Copyrighted Materials**

The study will include the use of secondary data and copyrighted materials; all necessary permissions will be obtained to ensure suitable and proper compliance. The sources used in the study will be appropriately cited to ensure transparency and integrity.

**K. Ethical Considerations for Animal and Human Trials**

The researcher will inform the human respondents (residents) involved in this study about the potential risks and benefits of their participation. They'll be told the study is safe and that only their responses and actions will be studied. However, if necessary, appropriate support services will be made available. No animal or other living organism will be involved in this study; therefore, ethical considerations are not applicable. The responses provided by the respondents in this study will not be disclosed to avoid any risk to their data security.

**L. Responsible Use of AI and Related Technologies**

The researchers will use AI but will ensure they do not rely solely on it, as it will only assist them in managing errors in the gathered data and in organizing their thoughts. For instance, the researchers will use ChatGPT in this study to help them better understand complex information. The tool will also help discover resources for the survey literature review, as it can assist in synthesizing broad data to identify the most relevant information for the study. The researchers will also use Grammarly to ensure the paper is grammatically correct and to check for possible plagiarism, in accordance with academic standards.

**M. Ethical Approval**

The study will be approved by the institutional review board of the ethics committee of Saint Michael College of Caraga. This validation ensures that we adhere to ethical guidelines designed to protect participants' privacy and intellectual property rights.

**Data Gathering Procedure**

The researchers will submit a letter of approval to the school head to obtain a study permit. The researchers will then coordinate with the school dean to secure approval to conduct the study in the selected locations. The study will involve senior high school students from Buenavista, Nasipit, and Carmen, Agusan del Norte. Additionally, the researchers will prepare the questionnaires and submit them to the research adviser for validation before data collection.

Afterward, the researchers will prepare a complete set of validated questionnaires. Before administering the questionnaires, an orientation will be conducted to explain the purpose of the study, procedures, potential risks, and the participants' rights, including the right to participate voluntarily and to withdraw at any time. The researchers will then distribute the questionnaires to the respondents in an actual field setting after securing the required informed consent and assent forms. Upon completion of the questionnaires, the researchers will collect all the survey responses. The data will then be compiled, analyzed, and interpreted accordingly.

**Letter of Approval.** The selected senior high school heads and administrators in Buenavista, Nasipit, and Carmen, Agusan del Norte, will receive a formal letter of request from the researchers. This letter requests approval to conduct the study and administer the research instruments to the students.

**Distribution of Informed Consent Form.** Respondents will receive informed consent forms, which serve as confirmation that they understand their rights and are willing to participate. For respondents under 18 years old, assent forms will be provided to parents or legal guardians, along with informed consent forms for the students. Participation will be allowed only after both forms have been duly signed and returned. For respondents aged 18 years and above, informed consent will be obtained directly from them. The informed consent forms will also inform participants that they may withdraw without adverse consequences.

**Distribution of Questionnaire**. The researcher will personally distribute the survey questionnaires to the respondents. Only respondents who have submitted the duly signed informed consent or assent form will be included in the survey. The researchers will allow respondents ample time to respond positively and honestly to the questions. All responses will be kept strictly confidential and used for academic purposes only.

**2.7 Research Data and Analysis Plan**

To gather precise and measurable information from the respondents, the researcher will use a quantitative descriptive-correlational research design to examine the extent of mobile banking utilization among senior high students in Buenavista, Nasipit, and Carmen, Agusan del Norte. The researchers will disseminate a survey questionnaire directly to qualified respondents as the primary method for data collection. The data will be collected, verified, tallied, presented, analyzed, and interpreted using the following statistical tools.

**Statistical Treatment**

The researchers will use the survey method to gather the data. The data will be interpreted, organized, and tabulated using the following statistical instruments.

**Percentage and Frequency.** It determined the respondent's demographic profile regarding age, sex, income (monthly allowance), and mobile banking platforms used.

**Weighted mean.** It addressed problems 2—the extent of mobile banking utilization among senior high students, including mobile banking features, perceived benefits, barriers, and proposed strategies.

**Pearson's r Correlation**. It answered problem 4, the relationship between the extent of mobile banking utilization and the respondent's demographic profile.

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**Membership/Affiliations**

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**Membership/Affiliations**

| ORGANIZATION | POSITIONS HELD |
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**Membership/Affiliations**

| ORGANIZATION | POSITIONS HELD |
|---|---|
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**Membership/Affiliations**

| ORGANIZATION | POSITIONS HELD |
|---|---|
| **Junior Financial Executive** | **MEMBER** |
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## APPENDICES

## CERTIFICATE OF AUTHENTIC AUTHORSHIP